# Fraud Detection: Method and Results

Oct 14, 2026 · Hasin Md. Daiyan

## Overview

The evidence layer, not the model architecture, carries the result: given the same past-only evidence, a fully neural hybrid is slightly below LightGBM on IEEE-CIS card fraud (0.630 vs 0.635 PR-AUC) and ahead of it on IBM AMLworld money laundering (0.650 vs 0.625), with the smallest seed-to-seed spread of any model on both.

The system has three parts.

- **Evidence layer.** 87 features built only from earlier transactions; a fraud label is used only once it is older than the label delay (30 days on IEEE-CIS, 1 day on AMLworld). Four parts: entity memory, deviation features, neighbour (graph) features and evidence signals.
- **Two neural component models.** TabM+G reads what the transaction looks like; the deviation model (+G) reads how it differs from the same user's past. Both receive the neighbour features.
- **Combination.** A fixed weighted average of the two scores, score = $w_1 \cdot s_1 + w_2 \cdot s_2 + b$, learned once and identical for every transaction. An evidence gate that sets the weights per transaction is reported as an ablation.

Baselines are LightGBM on raw features (trained once, and retrained every 30 days) and LightGBM+ (raw features plus the same evidence layer). All results are test-set means over seeds 42, 7 and 123 with paired bootstrap intervals, on IEEE-CIS and on IBM AMLworld HI-Small.

Code: `hybrid_ieee_cis.ipynb` (IEEE-CIS) and `hybrid_amlworld.ipynb` (AMLworld). Every number in this document is printed by one of these two notebooks or read from their stored per-seed result files.

## How the IEEE-CIS dataset is modified

No rows are removed and no values are invented: the data is merged, put in time order, given two derived identifier columns and split chronologically.

1. **Merge.** `train_transaction` is left-joined with `train_identity` on `TransactionID`. Transactions without identity data keep empty identity columns. Only the Kaggle training files are used, because the Kaggle test file has no labels. Result: 590,540 transactions.
2. **Time order.** Rows are sorted by `TransactionDT` (seconds from a reference point). Every later step depends on this order.
3. **First-use day.** D1n = TransactionDT // 86400 − D1, the day the card was first used.
4. **User identifier.** uid = card1 | addr1 | D1n. If any of the three parts is missing, the row gets no uid instead of being merged with other incomplete rows. 10.5% of test rows have no uid.
5. **Card bundle.** card_bundle = card1 | card2 | card3 | card5 | card4 | card6, one key for the full card description (built when card1 is present).
6. **Chronological split.**

| Split | Share | Rows | Fraud rate | Used for |
|---|---|---|---|---|
| Train | 70% | 413,378 | 3.52% | Training all models |
| Stop | 7.5% | 44,290 | 3.60% | Early stopping and model selection |
| Calibration | 7.5% | 44,291 | 3.27% | Platt and FCC calibration only |
| Test | 15% | 88,581 | 3.48% | All reported results |

Test rows by user familiarity: 33,166 with no earlier transaction (including rows without a uid), 30,387 with 1–5, and 25,028 with 6 or more.

In plain terms: the pipeline adds a stand-in for "one customer" (same card, same billing area, same card start date) and holds out the last 15% of the timeline, so every model is judged on transactions that happen after everything it learned from.

## Preprocessing

LightGBM receives the original columns almost untouched; the neural models receive scaled numeric columns, integer category codes and learned ID embeddings, all fitted on the training period only.

**LightGBM inputs.** All original columns except `TransactionID`, `isFraud`, `TransactionDT`, the split marker and the helper columns (`uid`, `card_bundle`, `D1n`). Text categories become integer codes, with −1 for a missing category. Missing numeric values stay

missing; LightGBM handles them natively.

**Neural inputs (401 numeric, 27 categorical columns).**

1. Numeric columns: signed logarithm sign(x)·log(1+|x|), standardised with training-period mean and standard deviation, clipped to ±8, missing set to 0.
2. Three added numeric columns: hour of day as sine and cosine, and the share of missing values in the row (these are part of the 401).
3. Categorical columns: integer codes with 0 for missing; a value never seen in training is also mapped to 0.

**Extra inputs for TabM+G (505 numeric, 36 categorical).**

- The evidence layer: entity memory (48), evidence signals (13), deviation features (11) and the 15 neighbour features.
- Frequency encodings: how often each of 12 identifier values (card1, card2, card3, card5, addr1, addr2, both email domains, DeviceInfo, id_31, card_bundle, uid) occurs in the training period (log count, standardised).
- Time encodings: weekly and monthly sine/cosine cycles, plus a linear time trend.
- ID embeddings: 9 identifier columns (card1, card2, card5, addr1, both email domains, DeviceInfo, id_31, card_bundle) get a learned 4-dimensional vector per value. Values seen fewer than 30 times in training share one "other" vector.
- Piecewise-linear encoding (PLE): each numeric column is cut into 8 bins at 9 quantile edges computed on 200,000 training rows. A value becomes 8 numbers saying how far it lies past each lower edge, which gives the network ready-made "x above threshold" steps, like a tree split.

In plain terms: LightGBM sees the data as Kaggle provides it. The neural models need numbers on a similar scale, so large values are compressed and centred, gaps are filled and every identifier gets a small learned description. Every statistic comes from the training period, so nothing about the test period leaks into the preparation.

## Feature engineering: the evidence layer

The evidence layer is 87 features computed from earlier transactions only, using a fraud label only once it is more than 30 days older than the transaction being scored. It is built before any model trains; LightGBM+ and both component models read the same layer.

| Part | Size | What it holds |
|---|---|---|
| Entity memory | 8 entities × 6 = 48 | For uid, card1, card_bundle, addr1, purchaser email, recipient email, DeviceInfo, id_31 (browser) |
| Evidence signals | 13 | How much is known about the transaction |
| Deviation features | 11 | How the transaction differs from the same user′s past |
| Neighbour (graph) features | 3 relations × 5 = 15 | Other users′ recent transactions on the same card1, addr1, DeviceInfo |

**Entity memory.** For each entity value, from earlier rows of that value only:

1. log number of earlier transactions;
2. log days since the value first appeared;
3. mean log amount of its earlier transactions;
4. smoothed fraud rate of its earlier transactions whose label is visible, (frauds + 10 × base rate) ÷ (visible labels + 10);
5. log number of visible labels;
6. first-sighting flag.

The smoothing pulls a value seen only a few times toward the overall fraud rate, so one fraud on a new card does not read as a 100% fraud rate.

**Evidence signals (13).** uid count, uid age, uid first-sighting flag, card1 count, D1, missing-value share, "no uid" flag, and, for each of the three relations, the number of recent neighbours found and the number of neighbours with a visible label. These describe how much is known, not what the transaction looks like.

**Deviation features (11).** Amount z-score against the user′s past amounts; log hours since the user′s last transaction; the user′s number of transactions in the last 1 hour, 24 hours and 7 days; and six first-time flags: first time this user shows this device, purchaser email, recipient email, browser, card bundle or product code. All are zero when the user has no earlier transactions.

**Neighbour (graph) features (15).** For each relation (card1, addr1, DeviceInfo), up to 8 most recent earlier transactions of other users with the same value, and up to 8 most recent whose label is already visible. Transactions of the same uid are excluded; those belong to

the deviation features. Per relation: log number of recent neighbours; log number of neighbours with a visible label; their smoothed fraud rate, (frauds + 2 × base rate) ÷ (visible labels + 2); mean log time gap to the recent neighbours; mean absolute log-amount difference to them.

**Leak checks (run once, stored).**

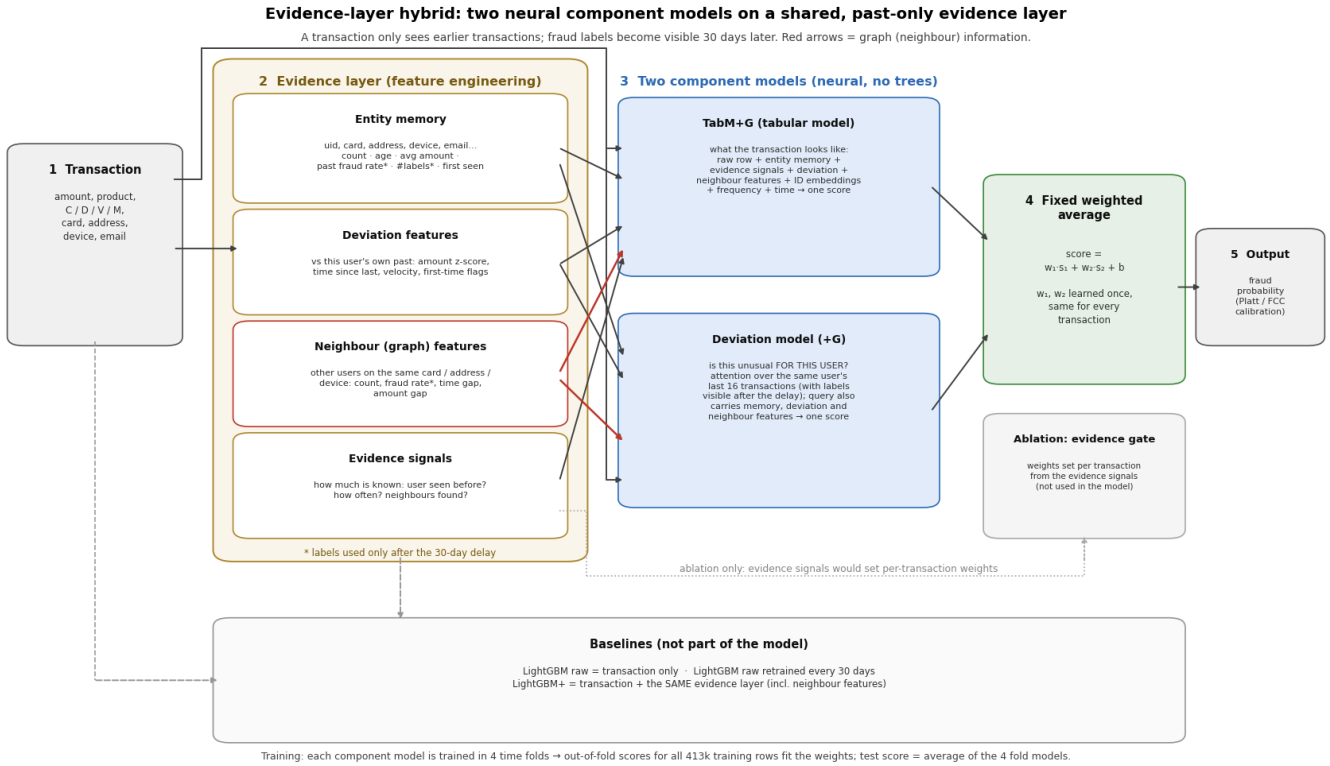
- Labels of the second half of the timeline are flipped and its amounts and categories changed; the entity memory (checked on uid), deviation and neighbour features of the first half do not change. All pass.
- Every user-sequence and neighbour index points only to earlier rows; label-visible neighbours are more than 30 days older. All pass.
- Null test: all labels are shuffled, the memory is rebuilt from the shuffled labels and a LightGBM (learning rate 0.1, 64 leaves, 200 rounds) is trained on the shuffled training labels. Test ROC-AUC against the shuffled labels is 0.505, 0.504 and 0.493 (mean 0.501), inside the chance band 0.5 ± 0.009.

In plain terms: for every transaction, the layer writes down what a fraud analyst would already know at that moment, using only the past, and fraud confirmations only after the 30 days a chargeback takes.

## Architecture and component models

One evidence layer feeds both component models and the LightGBM+ baseline; the neighbour features (red arrows) go into both models, and a fixed weighted average combines their two scores.

**Naming.** "+G" means the model also receives the 15 neighbour (graph) features. TabM+G and the deviation model (+G) are the component models of the hybrid; TabM+ and the deviation model without "+G" are the same networks without the neighbour features, used only as ablations.




The figure shows the IEEE-CIS setting (30-day label delay); on AMLworld the same architecture runs with a 1-day delay.

## Training scheme shared by both models

The training period is cut into 4 contiguous time blocks. For each block, one model is trained on the other three blocks, early-stopped on stop-split PR-AUC, and scores its held-out block plus the stop, calibration and test splits. This gives an out-of-fold score for every one of the 413,378 training rows, which the combination weights are fitted on. A test score is the average of the 4 fold models. Loss: binary cross-entropy (plus the L0 penalty where gates are used); optimiser AdamW (learning rate 0.001, batch 2,048, up to 30 epochs, patience 3).

The hybrid's test score therefore averages 8 networks (4 folds × 2 models), while each LightGBM baseline is one model; part of the hybrid's seed-to-seed stability comes from this averaging.

## TabM+G (tabular model)

- **Input:** 505 numeric columns (raw transaction, entity memory, evidence signals, deviation features, neighbour features, frequency and time encodings) and 36 categorical columns.

- **Feature selection:** one learned keep/drop gate per input column (hard-concrete L0, λ = 0.5), ramped in over the first 5 epochs; checkpoints are kept only after the ramp.
- **Encoding:** piecewise-linear encoding (8 bins) for each numeric column; a 4-dimensional embedding for each categorical column.
- **Network:** TabM-mini (Gorishniy et al., ICLR 2025). One MLP of width 256 with two hidden layers and dropout 0.1, shared by 16 members; each member multiplies the input by its own learned scaling vector and has its own output head. The score is the mean of the 16 member logits.

In plain terms: TabM+G judges what the transaction looks like, together with everything the evidence layer knows about its card, customer, address, device and neighbourhood, and 16 slightly different copies of it vote.

## Deviation model (+G)

- **Query (the current transaction):** its 19 core columns (amount, distance, C1, C2, C5, C6, C9, C11, C13, C14, D1, D2, D3, D4, D10, D15, hour sine and cosine, missing share), concatenated with the 11 deviation features, the 48 entity-memory features and the 15 neighbour features. A 2-layer network maps this to a 64-dimensional query.
- **Memory slots (the same user's last 16 transactions):** each slot holds that transaction's 19 columns, their difference from the current transaction, the log time gap and its label state: fraud, clean (older than 30 days) or pending. A 2-layer network maps each slot to 64 dimensions.
- **Empty slot:** a learned slot that is always present, so a user with no history still has something to attend to.
- **Attention and output:** 4-head attention from the query over the slots (dropout 0.1); the query and the attention summary are concatenated and a small network outputs one logit.

In plain terms: the model asks which of this customer's earlier transactions this one resembles, and how those turned out.

## Fixed weighted average (the combination used)

score = $w_1 \cdot s_1 + w_2 \cdot s_2 + b$, where $s_1$ and $s_2$ are the two models' logits and $w_1 + w_2 = 1$ (softmax of two learned numbers, so both positive). $w_1$, $w_2$ and b are learned once on the out-of-fold scores of all training rows, early-stopped on stop-split PR-AUC, and used for every transaction. TabM+G receives $w_1$ = 0.571, 0.487 and 0.499 on seeds 42, 7 and 123. Working on logits means the two models' odds are combined, not their probabilities.

### Evidence gate (ablation, not used)

The gate reads only the 13 evidence signals (never the transaction's own columns): a network 13 → 32 (ReLU) → 2 with a softmax gives per-transaction weights $w_1$ and $w_2$, and score = $w_1 \cdot s_1 + w_2 \cdot s_2 + b$, fitted like the fixed average. It has about 520 parameters against the fixed average's 3.

### Calibration

Scores are turned into probabilities with Platt scaling (a logistic fit on the calibration split) and with familiarity-conditioned calibration (FCC): a logistic fit on the score, the user's log transaction count, D1 and their products with the score, so the correction can differ between new and known users. Quality is measured as expected calibration error (ECE, 15 bins).

## Experimental protocol

Every model is trained and scored once per seed (42, 7, 123, fixed in advance and run in that order); results are test-set means over the three seeds with paired bootstrap intervals.

| Model | Inputs | Training |
|---|---|---|
| LightGBM raw | Raw transaction columns | Once on the training period, early-stopped on the stop split |
| LightGBM raw, retrained | Raw transaction columns | Retrained every 30 days from day 60 on all rows whose label is visible (older than 30 days), early-stopped on the last 15% of those rows; each model scores the next 30 days |
| LightGBM+ | Raw columns + the full evidence layer (87 features) | Once, as LightGBM raw |
| Hybrid | TabM+G + deviation model (+G), fixed weighted average | 4 time folds per component model; weights fitted on out-of-fold scores |

All LightGBM models use the same settings: learning rate 0.05, 256 leaves, minimum 100 rows per leaf, feature and row subsampling 0.5 and 0.8, L2 = 1, up to 3,000 rounds with early stopping after 100.

**Label delay.** 30 days is the main setting, chosen to match card chargeback timing. LightGBM+ without neighbour features is also run at 0, 7 and 60 days, once on seed 42.

**Metrics.**

- PR-AUC (average precision): the primary metric, because fraud is 3.5% of transactions.
- ROC-AUC.
- Recall at 1% flagged: the share of all frauds caught when the 1% highest-scoring transactions are reviewed (at most 0.29 here, since 1% of the test set is 886 transactions against about 3,080 frauds).
- By user familiarity: new (0 earlier transactions of the uid, including rows without a uid), few (1–5), regular (6 or more).
- Calibration: ECE.

**Significance.** Each comparison draws 200 bootstrap resamples of the test set and evaluates both models on the same resample. A difference is BETTER or WORSE only if its 95% interval excludes zero on every seed; otherwise it is a tie or mixed, and the per-seed verdicts are listed. The interval shown in the tables is the per-seed interval averaged over seeds, a summary rather than a confidence interval for the mean. The bootstrap covers test-set sampling, not training randomness; the three seeds cover the latter.

**Model selection.** The selection rule is to choose the hybrid variant with the highest stop-split PR-AUC and to prefer the simpler one on a tie. On IEEE-CIS the four variants differ by at most 0.0022 on the stop split (mean over seeds): no graph information 0.7485, graph features in both models 0.7476, graph as a third model 0.7472, graph features in TabM only 0.7463. The rule therefore selects the no-graph hybrid on IEEE-CIS. On AMLworld (seed 123, the only seed whose stop-split values are printed) the ranking is different again, and graph features in both models are ahead on test on every seed. Because the stop split cannot separate these variants and disagrees with the test split on both datasets, this document reports one architecture for both datasets, graph features in both models, and lists every variant. On IEEE-CIS the no-graph hybrid scores 0.6300 on test against 0.6296 for the reported model (a tie).

# IEEE-CIS results

The hybrid beats both raw LightGBM baselines on every seed (+0.084 and +0.109 PR-AUC) and is the most stable model (± 0.001). Against LightGBM+ with the same information it is 0.006 lower: worse on two seeds, level on one, with ROC-AUC equal. LightGBM+ is better for new and few-history users; the hybrid is better for regular users on every seed (+0.023).

Test set: 88,581 transactions, 3.48% fraud. Values are mean ± standard deviation over seeds 42, 7 and 123.

| Model | PR-AUC | ROC-AUC | Recall at 1% flagged |
|---|---|---|---|
| LightGBM, raw features | 0.5459 ± 0.0026 | 0.9041 ± 0.0017 | 0.257 |
| LightGBM, raw features, retrained every 30 days | 0.5203 ± 0.0071 | 0.8949 ± 0.0017 | 0.256 |
| LightGBM+ (raw + full evidence layer) | 0.6354 ± 0.0089 | 0.9344 ± 0.0013 | 0.270 |
| LightGBM+ without neighbour features | 0.6426 ± 0.0079 | 0.9351 ± 0.0014 | 0.272 |
| TabM+G alone | 0.6047 ± 0.0043 | 0.9300 ± 0.0021 | 0.263 |
| Deviation model (+G) alone | 0.5948 ± 0.0020 | 0.9282 ± 0.0010 | 0.266 |
| **Hybrid (TabM+G + deviation +G, fixed weighted average)** | **0.6296 ± 0.0013** | **0.9364 ± 0.0006** | 0.268 |

### PR-AUC by user familiarity

| Model | New (0) | Few (1–5) | Regular (6+) |
|---|---|---|---|
| LightGBM, raw features | 0.700 | 0.282 | 0.317 |
| LightGBM, raw, retrained every 30 days | 0.688 | 0.250 | 0.235 |
| LightGBM+ | 0.697 | 0.392 | 0.752 |

| Model | New (0) | Few (1–5) | Regular (6+) |
|---|---|---|---|
| LightGBM+ without neighbour features | 0.694 | 0.399 | 0.773 |
| Hybrid | 0.682 | 0.375 | 0.775 |

**What the hybrid beats** (hybrid minus the other model; verdicts per seed 42, 7, 123)

| Comparison | Δ PR-AUC | 95% interval | Verdict per seed | Δ ROC-AUC |
|---|---|---|---|---|
| LightGBM, raw features | +0.0837 | [+0.0704, +0.0969] | better, better, better | +0.0323 |
| LightGBM, raw, retrained every 30 days | +0.1092 | [+0.0951, +0.1231] | better, better, better | +0.0415 |
| LightGBM+ | −0.0058 | [−0.0127, +0.0015] | worse, worse, tie | +0.0019 |
| LightGBM+, new users | −0.0151 | [−0.0232, −0.0070] | worse, worse, worse | −0.0015 |
| LightGBM+, few-history users | −0.0180 | [−0.0331, −0.0061] | worse, tie, worse | −0.0044 |
| LightGBM+, regular users | +0.0229 | [+0.0108, +0.0343] | better, better, better | +0.0091 |
| LightGBM+ without neighbour features | −0.0131 | not computed | point differences −0.014, −0.005, −0.020 | +0.0013 |

**Which LightGBM+ is the reference.** LightGBM+ was defined as LightGBM with the full evidence layer, the same information the hybrid uses. On seed 123 the neighbour features hurt it (0.626 against 0.649 without them), which explains its larger spread and the tie on that seed. The stronger LightGBM+ without neighbour features is also reported, so the conclusion does not depend on that one seed.

**Finding.** A fully neural model given the evidence layer reaches 0.630 PR-AUC, 0.084 above raw LightGBM on every seed, with the smallest seed-to-seed spread of any model. Against LightGBM with the same information it is 0.006 lower (0.013 against the version

without neighbour features), with equal ROC-AUC. The user groups split consistently: trees are better where history is thin, and the hybrid is better for regular users on all three seeds.

## Analysis (IEEE-CIS)

The evidence layer, and specifically its delayed fraud labels, lifts performance on every seed; the architecture choices inside the hybrid move results by at most 0.02 PR-AUC, and only some of them are stable across seeds. Values are three-seed means; a result counts as established only when all three seeds agree.

### 1. What the evidence layer adds to LightGBM

| LightGBM inputs | PR-AUC | ROC-AUC |
|---|---|---|
| Raw features | 0.5459 | 0.9041 |
| Raw features, retrained every 30 days | 0.5203 | 0.8949 |
| + deviation features only | 0.5454 | 0.9059 |
| + evidence layer without any fraud labels | 0.5431 | 0.9083 |
| + entity memory only | 0.6403 | 0.9351 |
| + evidence layer without neighbour features | 0.6426 | 0.9351 |
| + full evidence layer (LightGBM+) | 0.6354 | 0.9344 |

| Comparison | Δ PR-AUC | 95% interval | Verdict per seed |
|---|---|---|---|
| LightGBM+ vs LightGBM raw | +0.0895 | [+0.0792, +0.1016] | better, better, better |
| LightGBM+ vs LightGBM raw retrained every 30 days | +0.1150 | [+0.1033, +0.1275] | better, better, better |
| Retrained every 30 days vs trained once | −0.0256 | [−0.0308, −0.0205] | worse, worse, worse |
| Entity memory only vs raw | +0.0944 | [+0.0835, +0.1058] | better, better, better |

| Comparison | Δ PR-AUC | 95% interval | Verdict per seed |
|---|---|---|---|
| Deviation features only vs raw | −0.0004 | [−0.0041, +0.0037] | better, tie, tie |
| Evidence layer without labels vs raw | −0.0028 | [−0.0079, +0.0027] | tie, worse, tie |
| Neighbour features added to LightGBM+ | −0.0073 | [−0.0111, −0.0032] | tie, better, worse |

On every seed the evidence layer lifts LightGBM by about +0.09 PR-AUC and beats a LightGBM retrained every 30 days by about +0.115. Almost all of the lift comes from the entity memory (+0.094), and within the memory from the delayed fraud labels: the same layer without labels adds nothing. Retraining on newly visible labels is worse than training once on all three seeds, so the gain is not explained by fresher labels. The deviation and neighbour features have no stable effect on LightGBM; the neighbour features changed it by −0.003, +0.004 and −0.023 on seeds 42, 7 and 123.

## 2. Label delay (LightGBM+ without neighbour features, seed 42)

| Label delay | PR-AUC | ROC-AUC | PR-AUC new users | PR-AUC few-history users | PR-AUC regular users |
|---|---|---|---|---|---|
| 0 days | 0.8294 | 0.9762 | 0.709 | 0.927 | 0.974 |
| 7 days | 0.7073 | 0.9542 | 0.703 | 0.538 | 0.916 |
| 30 days (main setting) | 0.6455 | 0.9361 | 0.699 | 0.404 | 0.770 |
| 60 days | 0.6098 | 0.9246 | 0.699 | 0.344 | 0.647 |

Performance depends strongly on how fast fraud is confirmed: 0.83 PR-AUC with immediate labels, 0.65 at 30 days. The effect is concentrated in known users; new users are unaffected (0.699–0.709), because they have no history to label. The 0-day row is an upper bound only, since it assumes a fraud is confirmed the moment it happens.

## 3. TabNet vs TabM (same inputs, same folds, no graph features)

| Model | PR-AUC | ROC-AUC | PR-AUC new users | Training time, 4 folds (min) |
|---|---|---|---|---|
| TabNet alone | 0.5179 ± 0.0196 | 0.9040 | 0.570 | 7.5 |
| TabM+ alone | 0.6025 ± 0.0041 | 0.9287 | 0.663 | 12.9 |
| Hybrid: TabNet + deviation (fixed average) | 0.6104 ± 0.0030 | 0.9305 | 0.672 | — |
| Hybrid: TabM+ + deviation (fixed average) | 0.6300 ± 0.0022 | 0.9356 | 0.685 | — |

TabNet vs TabM+: −0.0846 [−0.0955, −0.0746], worse on all three seeds. Inside the hybrid: −0.0196 [−0.0244, −0.0147], worse on all three seeds.

TabM is clearly the better tabular model: +0.085 PR-AUC alone and +0.020 inside the hybrid on every seed, at about 1.7 times TabNet's training time. TabNet is also about five times less stable across seeds (± 0.020 vs ± 0.004). It loses most on new users (0.570 vs 0.663), where many weak signals must be combined.

The structural reason: TabNet (Arik and Pfister, AAAI 2021) runs 5 sequential decision steps, each with a sparsemax mask that keeps a few features, so an unselected feature contributes nothing at that step (settings here: $n_d = n_a = 32$, $\gamma = 1.5$, sparsity weight $10^{-4}$, learning rate 0.02, ghost batch 512). TabM-mini keeps every feature available to all 16 members, prunes only through a learned per-feature gate, and gives each member threshold-like steps through the piecewise-linear encoding. Fraud evidence is spread across hundreds of weak C, D, V and identity columns plus the 87 evidence features, so looking at everything and averaging works better than picking a few.

## 4. Graph information: as a separate model or as features

All ways of using the graph land within 0.004 PR-AUC of each other on test; no single placement wins clearly, and the graph model as a separate (third) component adds a small gain on IEEE-CIS.

The graph model (ablation). For each transaction it takes, per relation (card1, addr1, DeviceInfo), the 8 most recent and the 8 most recent label-visible transactions of other users. Each neighbour is encoded from its own features plus its time gap and visible label. A learned keep/drop gate per neighbour (hard-concrete L0 relaxation, Louizos et al. 2018, λ = 0.02) prunes uninformative neighbours. The kept neighbours are averaged per set and a small network outputs one fraud score. Alone it scores 0.407 PR-AUC: 0.591 for new users but 0.056 for regular users.

| Model | PR-AUC | ROC-AUC | PR-AUC new | PR-AUC regular |
|---|---|---|---|---|
| Graph model alone | 0.4067 | 0.8514 | 0.591 | 0.056 |
| TabM+ → TabM+G | 0.6025 → 0.6047 | 0.9287 → 0.9300 | 0.663 → 0.662 | 0.725 → 0.738 |
| Deviation → deviation +G | 0.5995 → 0.5948 | 0.9299 → 0.9282 | 0.657 → 0.651 | 0.756 → 0.748 |

| Hybrid (fixed weighted average) | PR-AUC | ROC-AUC | PR-AUC new | PR-AUC regular | Stop-split PR-AUC |
|---|---|---|---|---|---|
| No graph information (TabM+ + deviation) | 0.6300 | 0.9356 | 0.685 | 0.770 | 0.7485 |
| Graph features in TabM only | 0.6310 | 0.9365 | 0.685 | 0.776 | 0.7463 |
| Graph features in both models (reported model) | 0.6296 | 0.9364 | 0.682 | 0.775 | 0.7476 |
| Graph as a third model (TabM+ + deviation + graph) | 0.6324 | 0.9354 | 0.687 | 0.766 | 0.7472 |
| Graph as features in both + third model | 0.6331 | 0.9363 | 0.684 | 0.772 | seed 123 only: 0.7504 |

| Comparison | Δ PR-AUC | 95% interval | Verdict per seed |
|---|---|---|---|
| Graph features added to TabM+ | +0.0022 | [−0.0001, +0.0045] | better, tie, worse |
| Graph features added to the deviation model | −0.0046 | [−0.0084, −0.0008] | tie, worse, tie |

| Comparison | Δ PR-AUC | 95% interval | Verdict per seed |
|---|---|---|---|
| Graph features in both models vs no graph | −0.0004 | [−0.0025, +0.0018] | better, worse, tie |
| Graph features in TabM only vs in both models | +0.0014 | [−0.0004, +0.0033] | tie, better, tie |
| Graph model added to the no-graph hybrid | +0.0025 | [+0.0008, +0.0044] | better, better, tie |
| Graph features in both models vs graph as a third model | −0.0029 | [−0.0057, −0.0002] | tie, worse, tie |
| Graph model added on top of graph features | +0.0035 | [+0.0016, +0.0056] | better, better, better |

Inside the deviation model the neighbour features help users with 1–5 earlier transactions (+0.010 on average; better, tie, tie) and cost regular users (−0.008; tie, worse, tie), whose own history is the stronger signal.

Why the effects are small. The entity memory already holds each card′s, address′s and device′s fraud rate over all users, so the cross-user signal the graph carries is largely present already. The graph model is informative mainly for new users (0.591), who have no history; for regular users its score is near chance (0.056). As a separate score it contributes one number that can only be weighted; as features it can interact with the transaction′s own details, but on IEEE-CIS that interaction does not add a measurable gain.

In plain terms: on card fraud, information about other customers on the same card, address or device moves the hybrid by less than 0.004 PR-AUC whichever way it is used. A separate graph model gives a small, consistent extra on IEEE-CIS (+0.0035 on top of the reported model, every seed); on AMLworld the same separate model hurts on every seed (section on AMLworld below), which is why it is not part of the reported architecture.

## 5. Evidence gate vs fixed weighted average

| Combination of TabM+G and deviation +G | PR-AUC | ROC-AUC | Δ PR-AUC vs fixed (95% interval) | Verdict per seed |
|---|---|---|---|---|
| Fixed weighted average (the model) | 0.6296 | 0.9364 | — | — |
| Evidence gate (ablation) | 0.6266 | 0.9353 | −0.0030 [−0.0042, −0.0017] | worse, worse, worse |

The gate is also worse for new users (−0.0038) and regular users (−0.0062) on every seed.

| User group | Evidence gate: TabM+G | Evidence gate: deviation +G | Fixed average: TabM+G | Fixed average: deviation +G |
|---|---|---|---|---|
| New | 0.637 | 0.363 | 0.519 | 0.481 |
| Few | 0.549 | 0.451 | 0.519 | 0.481 |
| Regular | 0.474 | 0.526 | 0.519 | 0.481 |

The fixed weights differ between seeds (TabM+G 0.571, 0.487 and 0.499), while the gate's pattern is the same on all three: it trusts TabM+G more for new customers and moves weight to the deviation model as the customer becomes known. Yet it scores below the fixed average on every seed. The reasons:

1. **The two models agree on most transactions.** Both score about 0.60 alone and read the same evidence layer, so re-weighting changes only the few transactions where they disagree. The fixed weights move between 0.49 and 0.57 across seeds with test PR-AUC staying at 0.629–0.631, which shows how flat this choice is.
2. **The gate's information is already inside the models.** The evidence signals are inputs to TabM+G, and the uid count and age are in the deviation model's entity memory, so each model has already adjusted its score for familiarity.
3. **A blend can only interpolate.** With positive weights that sum to one, the score lies between the two models' scores; it cannot create a pattern neither model found.
4. **Score-scale mismatch.** The weights are fitted on out-of-fold scores from single fold models, while test scores average four fold models and spread less. A per-transaction gate is more sensitive to this shift than one fixed blend.

5. **More parameters, drifting data.** About 520 parameters against 3, fitted on the earliest 70% of the timeline; per-group weights learned early fit the test period less well than one robust blend.

## 6. Why the neural models trail gradient-boosted trees given the same information

Given the same inputs, LightGBM+ scores 0.635–0.643 PR-AUC, TabM+G alone 0.605 and the hybrid 0.630. This matches the general finding on tabular data: tree ensembles remain the stronger default, which the FDB benchmark attributes to over-smoothing in neural networks, many uninformative features and the rotationally non-invariant nature of tabular data (Grover et al., citing Grinsztajn et al., 2022). On fraud data specifically:

1. **Sharp thresholds.** Fraud rules are often step-like (a D-column equal to 0, a card seen fewer than 3 times). A tree represents a step with one split; a network approximates it. Piecewise-linear encoding helps but does not close the gap.
2. **Many uninformative columns.** Trees ignore a column by never splitting on it; a network must learn to down-weight each one. TabM+G's L0 gates do this explicitly and still keep noise.
3. **Missing values carry signal.** Trees send missing values down their own branch; the networks see the training mean (0 after scaling) plus a missing-share column, which blurs "missing" with "average".
4. **High-cardinality identifiers.** Embeddings for rare card, address, email and device values are trained on few rows and stay noisy. This is where the hybrid loses most: new and few-history users.
5. **Few positives per pattern, and drift.** With 3.5% fraud and month-to-month change, each pattern has few examples; boosting fits them stage by stage with early stopping as regularisation.

Where the neural approach matches or beats trees is structured history: for regular users the hybrid scores 0.775 against 0.752 for LightGBM+ and 0.773 without neighbour features. The deviation model attends over a user's actual past transactions, a sequence a tree only sees as fixed summary features. The hybrid's PR-AUC also varies by ± 0.001 across seeds against ± 0.009 for LightGBM+, partly because each hybrid score averages 8 networks.

### 7. Calibration (ECE, lower is better)

| Model | Raw scores | Platt | FCC |
|---|---|---|---|
| Hybrid (fixed weighted average) | 0.0041 ± 0.0005 | 0.0024 ± 0.0002 | 0.0019 ± 0.0002 |
| LightGBM+ | 0.0046 ± 0.0012 | 0.0034 ± 0.0013 | 0.0040 ± 0.0022 |

After familiarity-conditioned calibration the hybrid's probabilities are the most reliable (ECE 0.0019; 0.0020, 0.0021 and 0.0016 on seeds 42, 7 and 123). LightGBM+ is less reliable and varies more, and FCC does not improve on Platt for it.

## Second dataset: IBM AMLworld HI-Small

On money laundering the hybrid beats LightGBM+ on every seed (+0.025 PR-AUC) and raw LightGBM by +0.49, and varies far less across seeds (± 0.003 vs ± 0.013). The same architecture, protocol and models were run on synthetic laundering transactions between bank accounts, to test whether the evidence layer and the neural hybrid hold up outside card fraud.

### How the two datasets differ

| | IEEE-CIS | AMLworld HI-Small |
|---|---|---|
| Domain | Card-not-present e-commerce fraud | Money laundering between bank accounts |
| Source | Real transactions (Vesta, Kaggle 2019) | Synthetic, agent-based simulator (Altman et al., NeurIPS 2023) |
| Rows used | 590,540 | 5,077,237 (8.6 times larger) |
| Period | About 6 months | 10 active days (1–10 Sept 2022) |
| Positive class | Fraud, 3.5% | Laundering, 0.089% (4,522 transactions; about 39 times rarer) |
| Customer identifier | Proxy: card1 + addr1 + card first-use day | Real: sending bank + account number |
| Raw columns | 430+ (C, D, M, V, identity, device, email) | 10 (amounts, currencies, payment format, banks, timestamp) |

| | IEEE-CIS | AMLworld HI-Small |
|---|---|---|
| Label delay (main) | 30 days | 1 day |
| Test rows with no history | 33,166 (37%; 10.5% of test rows have no uid at all) | 1,688 (0.2%) |

| Split | IEEE-CIS rows | IEEE-CIS fraud rate | AMLworld rows | AMLworld laundering (count, rate) | AMLworld days |
|---|---|---|---|---|---|
| Train | 413,378 | 3.52% | 3,554,065 | 2,856 (0.080%) | 0.00–6.62 |
| Stop | 44,290 | 3.60% | 380,793 | 348 (0.091%) | 6.62–7.40 |
| Calibration | 44,291 | 3.27% | 380,793 | 412 (0.108%) | 7.40–8.13 |
| Test | 88,581 | 3.48% | 761,586 | 906 (0.119%) | 8.13–10.00 |

**Rows removed from AMLworld.** The file spans 1–18 September, but 99.98% of transactions fall in 1–10 September. The remaining 1,108 rows (11–18 September) are about 59% laundering, an end-of-simulation artefact, and were removed so they cannot inflate test scores. No IEEE-CIS rows were removed.

AMLworld test rows by account familiarity: no earlier transactions 1,688; 1–5: 93,306; 6–20: 117,955; 21–100: 417,557; 101 or more: 131,080. Almost every test account has history, the opposite of IEEE-CIS.

## How the AMLworld code base differs

The AMLworld notebook is the IEEE-CIS final code base with a dataset adapter. The past-only protocol and leak checks, the evidence-layer functions, the component models, 4-fold blocked cross-fitting, the fixed weighted average and the gate ablation, the LightGBM settings and the seeds are identical. The paired bootstrap is the same 200 resamples per seed, reimplemented on the GPU as count-weighted sums that reproduce scikit-learn's PR-AUC and ROC-AUC. Only the parts below change.

| Part | IEEE-CIS | AMLworld | Why |
|---|---|---|---|
| Loading | Merge transaction and identity files; uid = card1 + addr1 + first-use day | Read the transactions file; uid = sending bank + account; owner entity joined from the accounts file | AMLworld has real account numbers and owners |

| Part | IEEE-CIS | AMLworld | Why |
|---|---|---|---|
| Entity memory (8 entities) | uid, card1, card bundle, addr1, both email domains, device, browser | Sending account, receiving account, sender→receiver pair, sending owner, receiving owner, sending bank, receiving bank, payment format + currencies | The entities that exist in each domain |
| Neighbour relations | Other users on the same card1, addr1, device | Other senders to the same receiving account and the same receiving owner, and other accounts of the same sending owner | Fan-in and shared ownership are laundering patterns |
| Deviation first-time flags | Device, both emails, browser, card bundle, product | Receiving account, receiving bank, payment format, both currencies, receiving owner | Same idea, domain columns |
| Sequence columns (deviation model) | 19 | 8 (paid and received amount, conversion ratio, self-transfer, same bank, currency mismatch, hour sine and cosine) | Columns that exist |
| Raw LightGBM features | All original columns | Amounts, currencies, payment format, bank IDs, flags, hour; account IDs excluded (keys, not features) | |
| D1 evidence signal | Card first-use day | Constant (no counterpart) | |
| Label delay | 30 days; sweep 0 / 7 / 60 | 1 day; sweep 0 / 2 / 3 | 10 active days; a 30-day delay would leave no label visible |
| Retrained baseline | Every 30 days from day 60 | Every day from day 3 | Scaled to the period |

| Part | IEEE-CIS | AMLworld | Why |
|---|---|---|---|
| Tail rows | None removed | 11–18 Sept removed | Simulation artefact |
| Pruning strength (L0) | TabM+ 0.5; graph model 0.02 | TabM+ 0.02; graph model 0.001 | See below |
| Hardware settings | LightGBM on CPU; neural models in 32-bit | LightGBM on GPU; neural models in mixed 16-bit precision | 8.6 times more rows; speed only |

**Pruning strength.** In a first AMLworld run, the IEEE-CIS pruning strength switched off almost every TabM+ input within five epochs and stop-split PR-AUC fell by half. With 0.08% positives the prediction loss per row is about 23 times smaller than with 3.5%, so the same penalty dominates training. The strength was therefore scaled by the base-rate entropy, the loss of a constant predictor:

$$\lambda_{\text{AML}} = \lambda_{\text{IEEE}} \cdot \frac{H(0.0008)}{H(0.035)} \approx 0.043\,\lambda_{\text{IEEE}}, \qquad H(p) = -p\ln p - (1-p)\ln(1-p)$$

This gives 0.02 for TabM+ and 0.001 for the graph model; all reported AMLworld results use it. Even so, TabM+'s gates still close to 31–42% open by the end of training (seed 123 log), so pruning remains strong on this dataset.

Leak checks on AMLworld: all past-only checks pass, and the null test gives ROC-AUC 0.503, 0.468 and 0.493 against shuffled labels (mean 0.488, inside the chance band 0.5 ± 0.017).

## AMLworld results

Test set: 761,586 transactions, 0.119% laundering. Means over seeds 42, 7 and 123. The LightGBM ablation rows marked "seeds 42, 123" were not run for seed 7.

| Model | PR-AUC | ROC-AUC | Recall at 1% flagged | Minority-class F1 (seed 123) |
|---|---|---|---|---|
| LightGBM, raw features | 0.1571 ± 0.0097 | 0.9673 | 0.490 | 0.230 |
| LightGBM, raw features, retrained daily (seeds 42, 123) | 0.1513 ± 0.0018 | 0.9673 | 0.481 | — |

| Model | PR-AUC | ROC-AUC | Recall at 1% flagged | Minority-class F1 (seed 123) |
|---|---|---|---|---|
| LightGBM+ (raw + full evidence layer) | 0.6253 ± 0.0133 | 0.9824 | 0.814 | 0.652 |
| LightGBM+ without neighbour features | 0.6169 ± 0.0084 | 0.9808 | 0.788 | 0.669 |
| TabM+G alone | 0.6375 ± 0.0020 | 0.9895 | 0.849 | 0.668 |
| Deviation model (+G) alone | 0.6327 ± 0.0018 | 0.9891 | 0.852 | 0.648 |
| **Hybrid (TabM+G + deviation +G, fixed weighted average)** | **0.6502 ± 0.0027** | **0.9905** | **0.855** | **0.675** |

Minority-class F1 uses the threshold that maximises F1 on the stop split, applied unchanged to the test split. It is printed for seed 123; seeds 42 and 7 need the F1 cell rerun.

**PR-AUC by account familiarity** (only 1,688 test rows belong to accounts with no history, so the "new" column has wide intervals)

| Model | New (0) | Few (1–5) | Regular (6+) |
|---|---|---|---|
| LightGBM, raw features | 0.075 | 0.161 | 0.167 |
| LightGBM+ | 0.116 | 0.564 | 0.681 |
| LightGBM+ without neighbour features | 0.100 | 0.561 | 0.672 |
| Hybrid | 0.172 | 0.593 | 0.699 |

| Comparison (hybrid minus …) | Δ PR-AUC | 95% interval | Verdict per seed | Δ ROC-AUC |
|---|---|---|---|---|
| LightGBM, raw features | +0.4931 | [+0.4619, +0.5203] | better, better, better | +0.0231 |
| LightGBM, raw, retrained daily (seeds 42, 123) | +0.4998 | [+0.4657, +0.5276] | better, better | +0.0233 |

| Comparison (hybrid minus …) | Δ PR-AUC | 95% interval | Verdict per seed | Δ ROC-AUC |
|---|---|---|---|---|
| LightGBM+ | +0.0249 | [+0.0118, +0.0366] | better, better, better | +0.0080 |
| LightGBM+ without neighbour features | +0.0333 | not computed | — | +0.0097 |
| LightGBM+, new accounts | +0.0564 | [−0.0193, +0.1987] | mixed | +0.0387 |
| LightGBM+, few-history accounts | +0.0291 | [+0.0085, +0.0481] | mixed | +0.0256 |
| LightGBM+, regular accounts | +0.0183 | [+0.0047, +0.0329] | mixed | +0.0034 |

## What the evidence layer adds to LightGBM (AMLworld)

| LightGBM inputs | PR-AUC | ROC-AUC |
|---|---|---|
| Raw features | 0.1571 | 0.9673 |
| Raw features, retrained daily (seeds 42, 123) | 0.1513 | 0.9673 |
| + deviation features only (seeds 42, 123) | 0.5279 | 0.9808 |
| + evidence layer without any labels (seeds 42, 123) | 0.5618 | 0.9827 |
| + entity memory only | 0.5443 | 0.9846 |
| + evidence layer without neighbour features | 0.6169 | 0.9808 |
| + full evidence layer (LightGBM+) | 0.6253 | 0.9824 |

| Comparison | Δ PR-AUC | 95% interval | Verdict |
|---|---|---|---|
| LightGBM+ vs LightGBM raw | +0.4682 | [+0.4369, +0.4965] | better on all seeds |
| LightGBM+ vs retrained daily (seeds 42, 123) | +0.4816 | [+0.4509, +0.5077] | better on both seeds |

| Comparison | Δ PR-AUC | 95% interval | Verdict |
|---|---|---|---|
| Retrained daily vs trained once (seeds 42, 123) | −0.0108 | [−0.0193, −0.0016] | mixed |
| Entity memory only vs raw | +0.3872 | [+0.3545, +0.4174] | better on all seeds |
| Deviation features only vs raw (seeds 42, 123) | +0.3658 | [+0.3349, +0.3946] | better on both seeds |
| Evidence layer without labels vs raw (seeds 42, 123) | +0.3997 | [+0.3640, +0.4312] | better on both seeds |
| Neighbour features added to LightGBM+ | +0.0084 | [−0.0043, +0.0220] | mixed |

**Label delay (LightGBM+ without neighbour features).** 0 days 0.6290, 2 days 0.5982, 3 days 0.6050 (seed 42); 1 day, the main setting, 0.6169 (mean of three seeds). Moving from immediate labels to a 2–3-day delay costs 0.02–0.03 PR-AUC.

## Design choices on AMLworld

| Comparison | Δ PR-AUC | 95% interval | Verdict |
|---|---|---|---|
| TabNet vs TabM+ (same inputs) | −0.0899 | [−0.1061, −0.0730] | worse on all seeds |
| Hybrid with TabNet vs with TabM+ | −0.0176 | [−0.0265, −0.0087] | mixed |
| Graph features added to TabM+ | +0.0111 | [+0.0069, +0.0154] | better on all seeds |
| Graph features added to the deviation model | +0.0101 | [+0.0021, +0.0185] | mixed |
| Graph features in both models vs no graph | +0.0096 | [+0.0053, +0.0142] | better on all seeds |
| Graph features in TabM only vs in both models | −0.0039 | [−0.0071, −0.0009] | mixed |

| Comparison | Δ PR-AUC | 95% interval | Verdict |
|---|---|---|---|
| Graph features in both models vs graph as a third model | +0.0150 | [+0.0094, +0.0206] | better on all seeds |
| Graph model added to the no-graph hybrid | −0.0055 | [−0.0085, −0.0024] | worse on all seeds |
| Graph model added on top of graph features | −0.0055 | [−0.0084, −0.0024] | worse on all seeds |
| Evidence gate vs fixed weighted average | −0.0005 | [−0.0037, +0.0023] | tie on all seeds |

The graph model alone scores 0.099 PR-AUC on every seed (about 80 times the 0.12% base rate, but far below the other models), and its edge gates end with 10–26% of edges open. Its rows therefore measure adding a weak model, not a tuned graph network. Calibration error is not informative here: with 0.12% positives almost every prediction falls in the lowest bin, and every model's ECE is below 0.0003.

## AMLworld compared with IEEE-CIS

| Question | IEEE-CIS | AMLworld |
|---|---|---|
| Evidence layer lifts LightGBM (LightGBM+ minus raw) | +0.090, better on all seeds | +0.468, better on all seeds |
| Memory vs retraining (LightGBM+ minus retrained raw) | +0.115, better on all seeds | +0.482, better on both seeds |
| Retraining vs training once | −0.026, worse on all seeds | −0.011, mixed |
| Lift without any labels | −0.003, no lift | +0.400, better on both seeds |
| Lift from the entity memory alone | +0.094 | +0.387 |
| Neighbour features added to LightGBM+ | −0.007, unstable sign | +0.008, unstable sign |
| Hybrid vs LightGBM+ | −0.006 (worse, worse, tie) | +0.025, better on all seeds |

| Question | IEEE-CIS | AMLworld |
|---|---|---|
| Hybrid vs LightGBM+, regular customers | +0.023, better on all seeds | +0.018, mixed |
| Hybrid vs LightGBM+, few-history customers | −0.018 (worse, tie, worse) | +0.029, mixed |
| Hybrid vs LightGBM+, new customers | −0.015, worse on all seeds | +0.056, mixed (1,688 rows) |
| TabM vs TabNet | +0.085, better on all seeds | +0.090, better on all seeds |
| Graph features in the hybrid (both models vs none) | −0.000, no stable effect | +0.010, better on all seeds |
| Graph model added on top of graph features | +0.0035, better on all seeds | −0.0055, worse on all seeds |
| Evidence gate vs fixed average | −0.003, worse on all seeds | −0.001, tie on all seeds |
| Hybrid seed-to-seed spread (PR-AUC sd) | ± 0.001 (LightGBM+ ± 0.009) | ± 0.003 (LightGBM+ ± 0.013) |

1. **The evidence layer** is the largest effect on both datasets, and retraining LightGBM never replaces it. What it captures differs. On IEEE-CIS the lift comes almost entirely from the delayed fraud labels; the label-free layer adds nothing. On AMLworld the label-free layer alone lifts LightGBM from 0.157 to 0.562 (seeds 42, 123), because simulated laundering follows fixed typologies (fan-in, fan-out, cycles, scatter-gather), so an account's counterparties, repeated pairs and velocity identify it before any label is visible.
2. **Label delay** matters much less where structure carries the signal: 0 to 30 days costs 0.18 PR-AUC on IEEE-CIS, 0 to 2–3 days costs 0.02–0.03 on AMLworld.
3. **Hybrid vs LightGBM+.** On IEEE-CIS the hybrid is slightly below LightGBM+; on AMLworld it is ahead on all three seeds, and ahead of LightGBM without neighbour features as well (+0.033). Two properties of AMLworld favour the neural models: real identities with long histories, where the deviation model is strongest, and only 10 raw columns, so the tree advantages of section 6 (sharp thresholds over many columns, missing-value signal, rare identifiers) matter less.

4. **Graph information.** On IEEE-CIS it is redundant with the entity memory: every placement lands within 0.004 PR-AUC. On AMLworld neighbour features improve TabM+ and the hybrid on every seed, because the relations encode laundering structure directly. The separate graph model goes the opposite way on the two datasets: a small gain on IEEE-CIS, a loss on AMLworld, every seed in both cases. Graph features in both models is the best placement on AMLworld (0.004 ahead of graph features in TabM only, mixed across seeds) and within 0.0035 of the best placement on IEEE-CIS, which is why it is the reported architecture.
5. **Design choices that transfer.** TabM beats TabNet on both datasets by almost the same margin (+0.085 and +0.090), and the evidence gate never beats the fixed average (worse on IEEE-CIS, tie on AMLworld).

**Overall finding.** The evidence layer is the dominant source of performance on both datasets, but what it captures depends on the domain: delayed fraud labels on card fraud, account structure on laundering. Given the same layer, the neural hybrid is slightly below LightGBM on IEEE-CIS and ahead of it on AMLworld on every seed. The advantage appears where the signal lies in account histories and account-to-account structure, which the deviation model and the neighbour features represent directly.

## How these results compare with published work

Published IEEE-CIS numbers are not directly comparable, because the split decides the score more than the model: random splits report ROC-AUC around 0.96 and PR-AUC up to 0.89, while out-of-time evaluations report ROC-AUC between 0.86 and 0.95. This work (strictly chronological, past-only features, 30-day label delay) reaches 0.934–0.936 ROC-AUC, in the upper part of the out-of-time range.

### IEEE-CIS

| Source | Evaluation | Model | ROC-AUC | PR-AUC |
|---|---|---|---|---|
| Kaggle competition, 1st place (NVIDIA blog) | Kaggle's later test file, private leaderboard | XGBoost + CatBoost + LightGBM ensemble with uid aggregates | 0.9459 | not reported |

| Source | Evaluation | Model | ROC-AUC | PR-AUC |
|---|---|---|---|---|
| Same team, local validation (NVIDIA blog) | First 75% train, last 25% validation | XGBoost, 216 features / with uid features | 0.9363 / 0.9472 | not reported |
| FDB benchmark (Grover et al.) | Out-of-time test, 67 features | LightGBM / CatBoost / MLP | 0.921 / 0.923 / 0.892 | not reported |
| FDB benchmark (Grover et al.) | Out-of-time test | Amazon Fraud Detector TFI / AutoGluon | 0.94 / 0.855 | not reported |
| Regulatory-framework study (arXiv 2605.04076) | 75/25 temporal split | XGBoost / LSTM | 0.9004 / 0.8579 | not reported |
| Ahmad et al. (2026), the reproduced paper | Their chronological split | TabNet / TabNet–GraphSAGE hybrid / LightGBM | not used here | 0.149 / 0.206 / 0.394 |
| LightGBM thesis (SDL repository) | Not out-of-time | Optuna-tuned LightGBM | 0.9606 | 0.8042 |
| Ensemble study (Tarif, Univ. Verona) | Not out-of-time; SMOTE variants | Stacking ensemble | 0.918 | 0.891 |
| This work (mean of seeds 42, 7, 123) | Chronological 70/15/15, past-only, 30-day label delay | LightGBM raw / LightGBM+ / hybrid | 0.904 / 0.934 / 0.936 | 0.546 / 0.635 / 0.630 |

The FDB authors show how much the split matters: AutoGluon reached 0.969 ROC-AUC in its own internal validation and 0.855 on the out-of-time test, a drop of 0.114, because its internal validation was not done out of time.

- ROC-AUC 0.934–0.936 for LightGBM+ and the hybrid is above the FDB out-of-time single models (0.92) and the 75/25 temporal XGBoost (0.90), level with the Kaggle winner's validation without uid features (0.936), and below the winning ensemble on Kaggle's test file (0.946), built from hundreds of engineered features and three boosting libraries.

- PR-AUC 0.630–0.635 is rarely reported under a time split. It is far above the reproduced paper (0.206 hybrid, 0.394 LightGBM) and below random-split reports (0.80–0.89), which are not comparable.
- Raw LightGBM at 0.904 ROC-AUC is close to the out-of-time LightGBM baselines in the literature (0.90–0.92), so the baseline is not weakened.

### AMLworld HI-Small

The AMLworld literature reports minority-class F1. The comparison is indicative, not head-to-head: the literature uses a temporal 60/20/20 split over all days while this work uses 70/7.5/7.5/15 and removes the 11–18 September tail; this work feeds past laundering labels into the entity memory after a 1-day delay, while the published graph models use no labels as inputs; and the published graph-pattern features are designed for the simulator's typologies, while the evidence layer here is generic and was designed on IEEE-CIS.

| Method | Minority-class F1 | Source |
|---|---|---|
| Gradient boosting, basic features only | at most 0.245 on the HI datasets | Blanuša et al., 2024 |
| GIN with multigraph adaptations | 0.572 | Egressy et al., AAAI 2024 |
| GFP + LightGBM / GFP + XGBoost (graph-pattern features) | 0.629 / 0.632 | Altman et al., NeurIPS 2023 |
| Multi-PNA with edge updates | 0.682 | Egressy et al., AAAI 2024 |
| Multi-FraudGT (graph transformer with reverse message passing, port numbering and ego IDs) | 0.761 | Lin et al., ICAIF 2024 |
| This work, seed 123: LightGBM raw / LightGBM+ / hybrid | 0.230 / 0.652 / 0.675 | this document |

Within those limits, raw LightGBM reproduces the published basic-feature result (0.230 vs at most 0.245), LightGBM+ with the generic evidence layer exceeds the purpose-built graph-pattern features (0.652 vs 0.629–0.632), and the hybrid lies just below Multi-PNA (0.675 vs 0.682).

## Other datasets (planned extension)

| Dataset | Standard evaluation | Reference results |
|---|---|---|
| European credit-card data (ccfraud, 0.17% fraud) | ROC-AUC, time-ordered test in FDB | 0.985–0.992 across AutoML frameworks (Grover et al.) |
| Elliptic Bitcoin (illicit transactions) | Illicit-class F1, 70:30 temporal split | Random Forest best in the original paper; GCN 0.628, Skip-GCN 0.705 (Weber et al., 2019) |

On ccfraud ROC-AUC is near its ceiling, so PR-AUC is the informative metric; Elliptic needs F1 alongside PR-AUC to be comparable.

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